

SUBNETTING EXAMPLE

- **Details & Requirements**
 - Network Address: 192.168.1.0
 - Default Subnet Mask: 255.255.255.0
 - Requires 2 Subnets
- **How many host bit do we need to borrow?**
 - 1 host bit, $2^1 = 2$ Subnets
- **How many addresses hosts per subnet?**
 - 7 host bits left, $2^7 = 128$ Addresses / Subnet
 - $2^7 - 2 = 126$ Addresses / Subnet
- **What are the valid subnets?**
 - 192.168.1.0 and 192.168.1.128
- **New Subnet Mask?**
 - 11111111.11111111.11111111.10000000
 - 255.255.255.128 or /25

Subnet	#1	#2
Network Address	192.168.1. 0	192.168.1. 128
First Host IP	192.168.1. 1	192.168.1. 129
Last Host IP	192.168.1. 126	192.168.1. 254
Broadcast Address	192.168.1. 127	192.168.1. 255

NETWORK ADDRESSING & SUBNETTING PRACTICE

TASK-1 (Network Addressing)

Do the calculations and find the appropriate values in *binary*, *decimal* or *hexadecimal* forms.

a. 89.232.34.118

- Binary

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- Hexadecimal

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b. 202.9.36.186

- Binary

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- Hexadecimal

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c. A9.36.05.DF

- Binary

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- Decimal

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d. 28.5B.29.87

- Binary

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- Decimal

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e. 01101100.11001011.00110110.01110110

- Hexadecimal

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- Decimal

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f. 10011011.00010011.11100010.00111011

- Hexadecimal

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- Decimal

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TASK-2 (Network Subnetting)

a. Do the calculations and find the appropriate values.

Details and Requirements

- Network Address : 192.168.100.0
- Default Subnet Mask : 255.255.255.0
- Subnets Required : 4 subnets

How many **host bits** do we need to borrow?

-

How many **host addresses** per subnet?

-

-

What are the **valid subnets**?

-

New **Subnet Mask**?

-

-

Subnet	#1	#2	#3	#4
Network Addr.				
First Host IP				
Last Host IP				
Broadcast Addr.				

